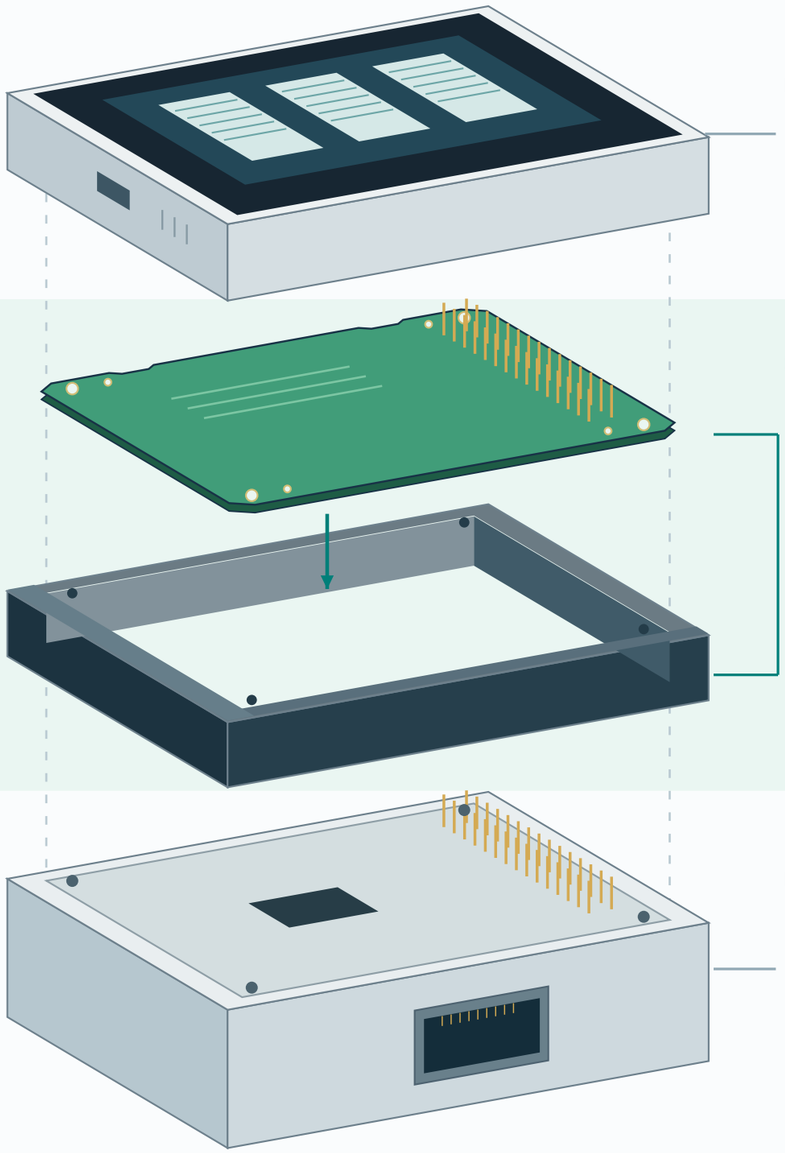
Final assembly: the M5Stack shade bridge

Three enclosure layers. The custom radio carrier lives inside the middle M032 housing.

EXPLODED VIEW / TOP TO BOTTOM



1

CoreS3 SE

M5Stack K128-SE

ESP32-S3 compute + touchscreen
USB-C power enters this layer.

2

M032 radio layer

Custom carrier PCB - lifted out for clarity

50 x 50 mm carrier replaces the perfboard.
Radio and components face DOWN inside.
M5Stack A089 socket is part of this PCBA.

Module13.2 Proto housing

M5Stack M032 - retain this frame

The board and frame form ONE middle layer.

3

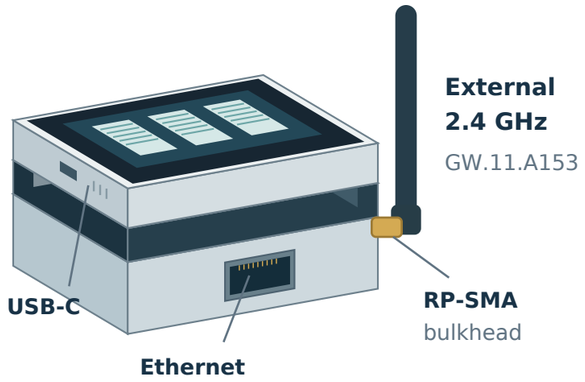
Base LAN v1.2

M5Stack K012-B-V12

W5500 Ethernet + bottom enclosure
RJ45 network connection enters here.
No additional battery-bottom layer.

30-pin M-Bus passes through the carrier between Core and LAN base.

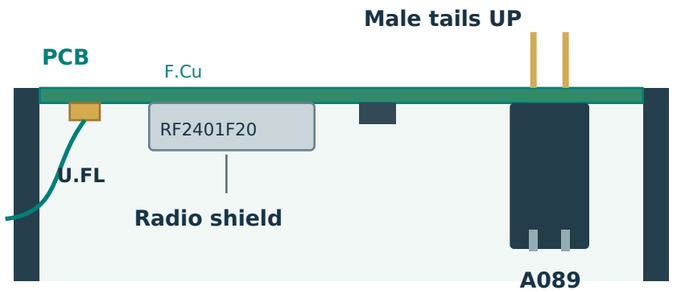
ASSEMBLED APPLIANCE



Core / M032 / LAN are the three case layers.
Views simplify ports and antenna placement.

INSIDE THE M032

Simplified side section - Core above, LAN below



All SMT parts face down into the ring.
A089 female socket faces the LAN base.
Coax stays inside; antenna is outside.

INSTALLATION DETAILS

Keep the M032 frame

Remove its original perforated PCB. Reuse its four M2 x 5 mm self-tapping PCB retention screws.

Internal cable + external antenna

CAB.628: U.FL-compatible plug to RP-SMA bulkhead.
Add strain relief; mount antenna to the bulkhead.

First-article fit details

Confirm pin engagement, stack screw lengths and bulkhead hole location on an actual sample.